

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech. Examination (Mechanical-CAD/CAM)****May 2018****ME766 Computational Fluid Dynamics****Date: 10.05.2018, Thursday Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION- I**Q.1 Attempt the following**

- (a) Experimental testing stage can be reduced, but cannot be completely eliminated using CFD. Comment. **[05]**
- (b) List down the commercially available CFD packages available in the market. **[02]**

Q.2 Attempt any two of the following**[14]**

- (a) A two-dimensional small-disturbance velocity potential equation for compressible flows is given as:

$$(1 - M_{\infty}^2) \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

where M is the Mach number of flow. (i) Examine whether this equation is parabolic, elliptic, or hyperbolic? (ii) Justify your inference from pure physical arguments.

- (b) State Reynolds's Transport theorem. Derive a Continuity equation by using Reynolds's Transport theorem
- (c) Using Taylor's series derive second order central-difference for $\frac{\partial u}{\partial x}$.

Q.3 Attempt any two of the following**[14]**

- (a) Explain the various components of Numerical Solution and errors in CFD tools.

- (b) Steady state temperature distribution in a rectangular plate is governed by the Laplace equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \text{ in } x \in (0,1), y \in (0,1)$$

subject to the boundary conditions $T(0,y) = T(1,y) = T(x,0) = 0$ and $T(x,1) = \sin \pi x$ Prepare the difference equation.

- (c) Explain the Explicit and Implicit Approach for transient problem with its advantages and limitations.

SECTION- II

Q.4 Attempt the following

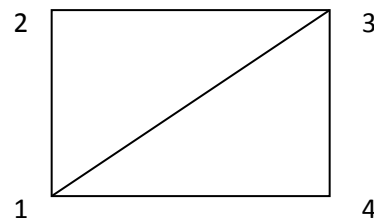
- (a) Explain the polynomial approach to find a second order accurate finite difference at the boundary. [05]
- (b) Why is FEM termed as “piecewise approximation”? [02]

Q.5 Attempt any two of the following

[14]

- (a) Explain structured grid and Unstructured grid.
- (b) Find the temperature at point P by FEM.

Point	X-coordinate	Y-Coordinate
1	0	0
2	0	1
3	1	1
4	1	0
P	MM / 20	DD / 35



$T_1 = DD$, $T_2 = 2 * DD$, $T_3 = \sqrt{YYYY}$, $T_4 = MM$ Where DD- You're Birth Date, MM: - Your Birth Month and YYYY- is Your Birth Year.

- (c) Consider the solution of the following template 1 –D wave equation:

$$\frac{\partial u}{\partial t} + C \frac{\partial u}{\partial x} = 0$$

Using a modified FTCS scheme, in which the term u_i^n for time discretization is expressed as $u_i^n = \frac{1}{2}(u_{i+1}^n + u_{i-1}^n)$, where the index ‘i’ represents spatial discretization whereas the superscript ‘n’ represents temporal discretization. Examine the numerical stability of this scheme using von-Neumann stability analysis.

Q.6

Attempt any two of the following

[14]

- (a) Explain the Various Convection Schemes (First order Upwind Scheme, Central difference Scheme, Quick Scheme) used in Finite volume method.
- (b) Explain the physical meaning of Elliptical, Parabolic and Hyperbolic equation with suitable example.
- (c) i) Explain what is post-processing and pre-processing in CFD.
ii) Explain the elliptic grid generation.